

Logical Operators Quick Reference				
$\neg$	$\wedge$	$\vee$	$\rightarrow$	$\leftrightarrow$

1A. If you study for the exams, then you will do well in discrete math.

1B. If you remember to bring money, then you can go out to eat.

1C. If you do well on the exams, then you will pass the course.

2A. True

2B. True

2C. True

2D. False

3A. If you are taking discrete math and it is snowing, then you are a student.

3B. It is not snowing or if it is snowing then you are a student.

3C. It is snowing and you are taking discrete math if and only if you are a student.

4A. $l \rightarrow (\neg a \vee \neg h)$

$\neg (l \rightarrow (\neg a \vee \neg h))$	Negation
$\neg (\neg l \vee (\neg a \vee \neg h))$	Conjunctive Disjunction
$l \wedge \neg (\neg a \vee \neg h)$	DeMorgan's Laws
$l \wedge a \wedge h$	DeMorgan's Laws
You are late to class and you set an alarm and you did your homework.	Translation

4B. $a \rightarrow \neg l$

$\neg (a \rightarrow \neg l)$	Negation
$\neg (\neg a \vee \neg l)$	Conjunctive Disjunction
$a \wedge \neg \neg l$	DeMorgan's Laws
$a \wedge l$	Double Negation Laws
You set an alarm and you are late for class.	Translation

5. Statement: I go to bed whenever I finish my school project.

Converse: Whenever I finish my school project, I go to bed.

Inverse: I do not go to bed when I do not finish my school project.

Contrapositive: When I do not finish my school project, I do not go to bed.

6A. $\neg p \rightarrow \neg q$

p	q	$\neg p$	$\neg q$	$\neg p \rightarrow \neg q$
T	T	F	F	T
F	T	T	F	F
T	F	F	T	T
F	F	T	T	T

6B. $(\neg p \wedge q) \rightarrow \neg r$

p	q	r	$\neg p$	$\neg p \wedge q$	$\neg r$	$(\neg p \wedge q) \rightarrow \neg r$
T	T	T	F	F	F	T
F	T	T	T	T	F	F
T	F	T	F	F	F	T
F	F	T	T	F	F	T
T	T	F	F	F	T	T
F	T	F	T	T	T	T
T	F	F	F	F	T	T
F	F	F	T	F	T	T

6C. $(p \rightarrow \neg q) \leftrightarrow \neg(p \vee q)$

p	q	$\neg q$	$p \rightarrow \neg q$	$(p \vee q)$	$\neg(p \vee q)$	$(p \rightarrow \neg q) \leftrightarrow \neg(p \vee q)$
T	T	F	F	T	F	T
F	T	F	T	T	F	F
T	F	T	T	T	F	F
F	F	T	T	F	T	T

7A.

$q \rightarrow (p \vee q)$	Given
$\neg q \vee (p \vee q)$	Conjunctive Disjunction
$\neg q \vee p \vee q$	Associative Laws
$T \vee p$	Negation Laws
T	Identity Laws

7B.

$(p \rightarrow q) \wedge (p \rightarrow \neg q)$	Given
$(\neg p \vee q) \wedge (\neg p \vee \neg q)$	Conjunctive Disjunction
$q \vee (\neg p \wedge p)$	Distributive Laws
$q \vee F$	Negation Laws
q	Identity Laws

8A.

p	q	$(p \wedge q)$	$(p \wedge q) \rightarrow q$
T	T	T	T
F	T	F	T
T	F	F	T
F	F	F	T

p	q	$\neg q$	$p \wedge \neg q$	$(p \wedge \neg q) \rightarrow \neg q$
T	T	F	F	T
F	T	F	F	T
T	F	T	T	T
F	F	T	F	T

Q.E.D.

8B.

$(p \wedge q) \rightarrow q$	Given
$\neg (p \wedge q) \vee q$	Conjunctive Disjunction
$(\neg p \vee \neg q) \vee q$	DeMorgan's Laws
$\neg p \vee (\neg q \vee q)$	Associative Laws
$\neg p \vee T$	Negation Laws
T	Domination Laws

$(p \wedge \neg q) \rightarrow \neg q$	Given
$\neg (p \wedge \neg q) \vee \neg q$	Conjunctive Disjunction
$(\neg p \vee \neg \neg q) \vee \neg q$	DeMorgan's Laws
$(\neg p \vee q) \vee \neg q$	Double Negation Law
$\neg p \vee (q \vee \neg q)$	Associative Laws
$\neg p \vee T$	Negation Laws
T	Domination Laws

Q.E.D.

9.

A	B	C	$\neg B$	$A \vee \neg B$	$\neg A$	$C \rightarrow \neg A$	$\neg C$	$\neg C \vee \neg A$	$B \rightarrow (\neg C \vee \neg A)$
T	T	T	F	T	F	F	F	F	F
F	T	T	F	F	T	T	F	T	T
T	F	T	T	T	F	F	F	F	T
F	F	T	T	T	T	T	F	T	T
T	T	F	F	T	F	T	T	T	T
F	T	F	F	F	T	T	T	T	T
T	F	F	T	T	F	T	T	T	T
F	F	F	T	T	T	T	T	T	T

The truth table reveals there are two instances in which the initial truth values of A, B, and C correspond to the logical statements provided by each person. Therefore, since between each iteration Person A and Person B's initial truth values differ, then their status cannot be determined. However, Person C's initial truth corresponds to their logical statement value in both instances, so they are a Viking.

Regardless of if Person C's statement were to be interpreted as $\neg C \vee (B \rightarrow \neg A)$, the logical statement value is equivalent to the one provided in the table, thus providing the same result.